

NovaCAD - Implemented Features & Requirements

Core Viewer Capabilities

- **DXF/DWG file loading** — supports .dxf, .dwg (via ODA File Converter or LibreDWG), .zip (eTransmit packages), and folders of drawings
- **AutoCAD-comparable rendering** — per-entity/per-layer colors (full ACI palette + true color), dark/light canvas, linetypes (dashed/center/hidden), hatches, solids, dimensions, text with alignment/rotation
- **Entity support** — LINE, CIRCLE, ARC, LWPOLYLINE (with bulge arcs), POLYLINE (2D/3D, polyface meshes, grids), SPLINE (NURBS), ELLIPSE, SOLID, TRACE, 3DFACE, POINT, HATCH, TEXT, MTEXT, ATTRIB, INSERT (with arrays), DIMENSION, LEADER, ACAD_TABLE
- **AutoCAD semantics** — block entities on layer 0 inherit insert's layer, BYLAYER/BYBLOCK colors/linetypes resolve correctly, frozen/off layers start hidden
- **Performance optimization** — async bitmap render pipeline (30–80 ms redraws on 3.4M-entity files), byte-level streaming parser (no full-file string conversion), render-time LOD via point-run decimation bounded by screen area
- **Huge file support** — handles multi-million-entity drawings; tested on 731 MB / 3.4M-entity plant layout

Search & Navigation

- **Deep search (§F)** — Spotlight-style search over text, labels, attributes, block names, and xref content; results update live; Enter/arrows step through matches with animated pan/zoom and 1.5s selection halo
- **Zoom controls** — trackpad pinch zoom, two-finger scroll pans, Option+scroll, mouse wheel, drag panning
- **Fit-to-view** — initial fit uses robust (percentile) extents; press Fit twice for true full extents

Measurement Tools

- **Distance mode** — click two points → distance, $\Delta X/\Delta Y$, angle
- **Area mode** — click boundary points, double-click/Enter to close → area + perimeter
- **Radius mode** — pick circle/arc → radius readout
- **Angle mode** — 3-point → angle measurement
- **Live preview** — rubber-band preview with live readout chip
- **Unit systems** — parsed from \$INSUNITS; display formats: decimal, architectural, engineering, fractional, scientific ($\times$ imperial/metric/as-drawn $\times$ precision)

Drawing Tools & Markup

- **Drawing tools** — Line, Polyline, Circle, 3-point Arc, Rectangle, Polygon, Text, Erase on a NOVACAD-MARKUP layer
- **OSNAP** — endpoint, midpoint, center, intersection, perpendicular, tangent (with visual markers)
- **Typed input** — absolute (100,50), relative (@25,0), length along cursor direction, polygon side count, close polyline
- **Markup color** — selectable from toolbar (red, green, blue, ...), saved across launches, travels with exported/merged DXF
- **Command palette** — AutoCAD-style command line at bottom: L , PL , C , A , REC , POL , T , E , DI , AREA , Z , U

Edit Markup

- **Select markup** — click to select (Shift extends, Esc deselects)
- **Move markup** — drag selected entities
- **Delete markup** — Delete key removes selected markup
- **Recolor markup** — change color via popup menu
- **Edit properties** — editable Markup panel shows type, layer, color
- **Undo** — ⌘Z removes last action (grouped for stamps)

Markup Export & Merging

- **Export Markup as DXF** — standalone DXF for use as xref over master drawing in AutoCAD
- **Save Copy with Markup** — byte-faithful copy of original DXF with markup entities merged in (NOVACAD-MARKUP layer injected into LAYER table + entities)
- Both preserve original drawing content untouched

Block Operations

- **Stamp blocks** — pick existing block symbol, click to place copies as markup with ghost preview and OSNAP
- **Top-250 blocks** — up to 250 block symbols captured (limited to ≤200 primitives each, ≤40K total)
- **Explode on stamp** — symbol explodes to markup lines (not INSERT references)

Layer Management

- **Layer panel** — show/hide (freeze/thaw) any layer, color swatches, entity counts, search filter
- **Lock/unlock layers** — locked layers stay visible but can't be selected or snapped
- **Bulk controls** — All On / All Off buttons
- **Isolate layer** — double-click to isolate
- **Xref layer naming** — AutoCAD-style `XREFNAME|layer` in layer panel

Xref & Package Support

- **eTransmit packages** — open .zip directly, extracts and resolves all xrefs (nested/circular)
- **Folder/file opens** — xrefs resolve against sibling files automatically
- **Xref panel** — show/hide each external reference independently
- **Xref detection** — missing/unresolved xrefs flagged (e.g., external `P:\...` DWG paths with no geometry)
- **DWG converter stack** — ODA File Converter (preferred) or GNU LibreDWG fallback (DXF members always work)
- **eTransmit report parsing** — auto-detects main drawing via package's .txt report

Selection & Properties

- **Click selection** — click objects to select (block insert selects whole block reference, like AutoCAD)
- **Multi-selection** — clicks accumulate (AutoCAD PICKADD), Shift+click removes, Esc/clicking empty space deselects
- **Properties panel** — type, layer, color, linetype, geometry (length, area, radius, text contents, block name/scale/rotation, ...); minimizable to edge tab
- **Multi-selection merge** — "Various/Multiple" shown for differing properties

Document Operations

- **Reload** — re-open drawing and xrefs from disk, preserving view, markup, and layer state
- **Quality preference** — 1–5 scale (Fastest → Superb) saved across launches; Reset returns to 3
- **Unsaved-markup warning** — confirm on open if markup exists

Headless Snapshots

- **CLI rendering** — `--snapshot out.png` renders PNG without UI
- **Options** — `--size` , `--light` , `--space paper` , `--focus x,y,w,h` , `--select x,y` , `--quality` , `--debug-bounds`
- **Batch preview** — works on .dxf, .dwg, .zip, folders

Application Packaging

- **macOS app bundle** — `Scripts/build_app.sh` compiles, code-signs, generates icon, creates Info.plist
- **Document type registration** — .dxf/.dwg files open with NovaCAD in Finder/Launchpad
- **Re-run after code changes** — script refreshes installed app

Deferred/Not Implemented (Documented Rationale)

- **Metal/3D rendering** — current CoreGraphics pipeline meets perf goals; Metal remains optional for 3D/orbit
- **DWG writing** — only DXF export; requires licensed ODA SDK for native DWG
- **Original entity editing** — editing is additive markup only; source geometry never modified
- **Block INSERT writing** — Stamp tool explodes blocks to lines, not INSERT entities
- **Plot settings** — not exposed (Lock/Freeze are)